

Normal Distribution

If X is a continuous random variable following normal probability distribution with mean μ and standard deviation σ , then its probability density function (p.d.f) is given by

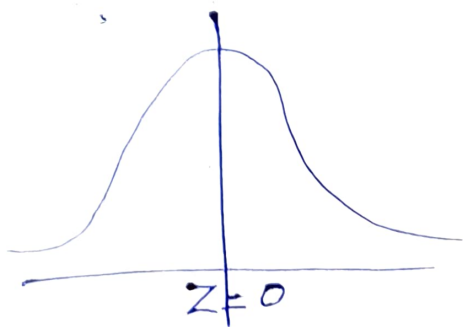
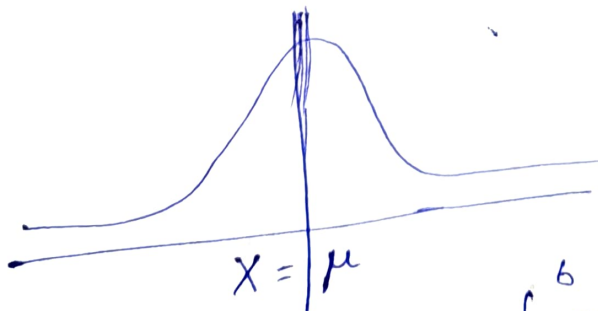
$$P(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

• Standard Normal Distribution:

If X is a random variable following normal distribution with mean μ & standard deviation σ , then the random variable Z is defined as

$$Z = \frac{X - E(X)}{\sigma_x} = \frac{X - \mu}{\sigma}$$

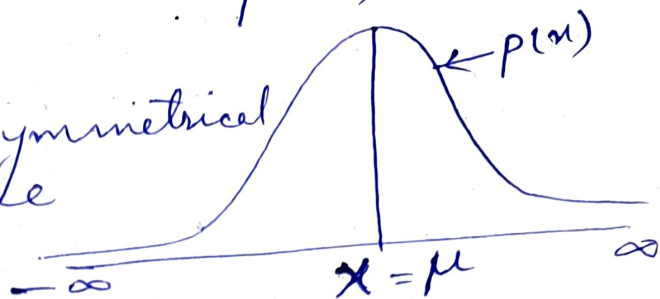
is called standard normal variate
Here $\mu = 0$ & $\sigma = 1$



$$P(a \leq x \leq b) = \int_a^b P(x) dx$$

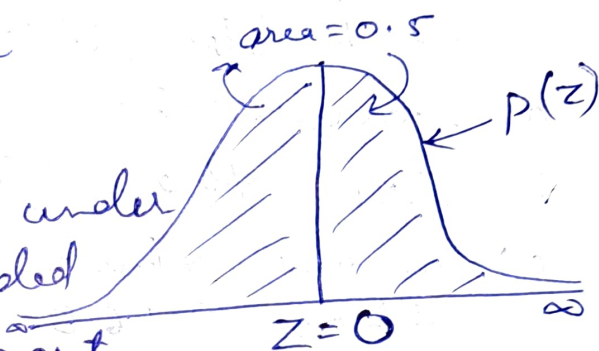
- The normal curve $p(x)$ is bell shaped

- The curve is symmetrical about the line $X = \mu$ ($Z = 0$)



- Mean = Median = Mode = μ

- The whole area under the curve is divided into two equal parts.



- Since total area under normal probability curve is 1, area to the right of $X = \mu$ is equal to the area to the left of $X = \mu$ & is equal to 50% (0.5).

- No portion of the curve lies below x -axis, since $p(x)$ can never be negative.

- Moments of odd order about the mean is zero
i.e., $\mu_{2n+1} = 0$, $n = 0, 1, 2, \dots$

i.e., $\mu_1 = \mu_3 = \mu_5 = \dots = 0$

- Moments of even order about the mean is given by
 $\mu_{2n} = 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1) \sigma^{2n}$
 $n = 1, 2, 3, \dots$

i.e., $\mu_2 = \sigma^2$, $\mu_4 = 1 \cdot 3 \sigma^4 = 3\sigma^4$

TABLE VI: AREAS UNDER STANDARD NORMAL PROBABILITY CURVE

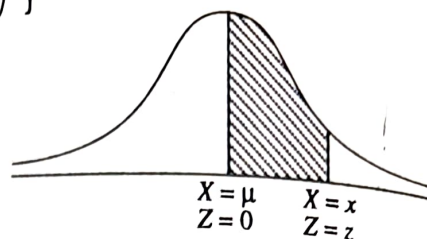
Normal Probability curve is given by : $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right\}; -\infty < x < \infty$

and standard normal probability curve is given by :

$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right), -\infty < z < \infty$$

where

$$Z = \frac{X - E(X)}{\sigma_x} = \frac{X - \mu}{\sigma} \sim N(0, 1)$$



The following table gives the shaded area in the diagram, viz. $P(0 < Z < z)$ for different values of z .

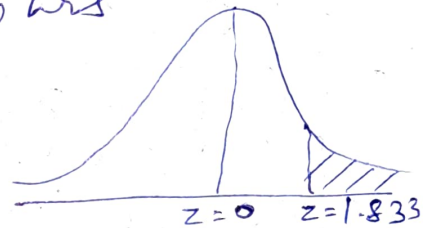
AREAS UNDER STANDARD NORMAL PROBABILITY CURVE

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.9	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

Q:- In a test on 2000 electric bulbs, it was found that the life of a particular make, was normally distributed with an average of 2040 hours & S.D of 60 hours. Estimate the no. of bulbs likely to burn for

- more than 2150 hours
- less than 1950 hours
- more than 1920 hrs but less than 2160 hours.

Solⁿ:- Here $\mu = 2040 \text{ hrs}$
 $\sigma = 60 \text{ hrs}$



(a) For $X = 2150$

$$Z = \frac{X - \mu}{\sigma} = 1.833$$

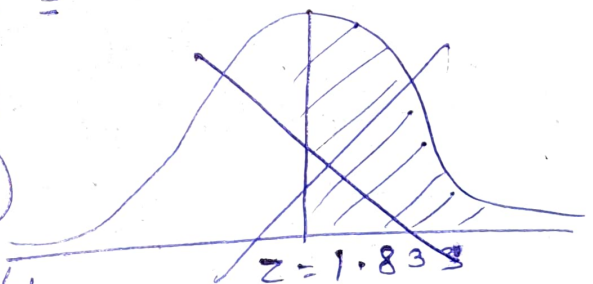
$$\therefore P(X > 2150)$$

$$= P(Z > 1.833)$$

$$= 0.5 - 0.4664$$

$$= 0.0336$$

$$\begin{aligned} \text{No. of bulbs} &= 0.0336 \times 2000 \\ &= 67 \text{ (approx).} \end{aligned}$$

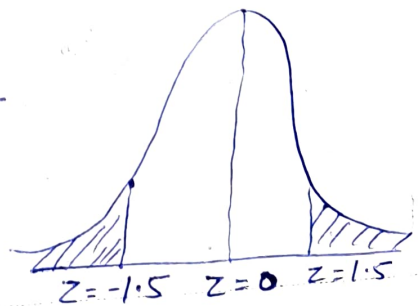


(b) For $X = 1950$

$$Z = \frac{X - \mu}{\sigma} = -1.5$$

$$\therefore P(X < 1950)$$

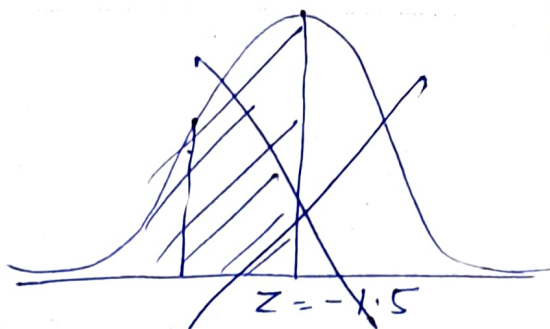
$$= P(Z < -1.5)$$



$$\begin{aligned}
 &= P(Z > 1.5) \\
 &= 0.5 - 0.4332 \\
 &= 0.0668
 \end{aligned}$$

No. of bulbs

$$\begin{aligned}
 &= 0.0668 \times 2000 \\
 &\approx 134 \text{ (approx).}
 \end{aligned}$$



(C) For $X = 1920$, $Z = \frac{X - \mu}{\sigma} = -2$

For $X = 2160$, $Z = \frac{X - \mu}{\sigma} = 2$

$$\therefore P(1920 < X < 2160)$$

$$= P(-2 < Z < 2)$$

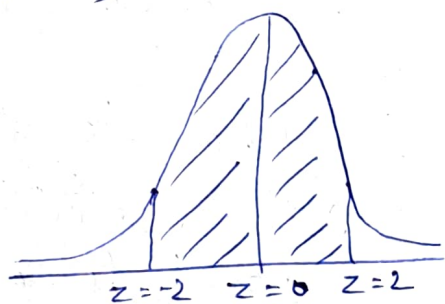
$$= P(-2 < Z < 0) + P(0 < Z < 2)$$

$$= 2 P(0 < Z < 2)$$

$$= 2 \times 0.4772$$

$$= 0.9544$$

$$\begin{aligned}
 \text{No. of bulbs} &= 0.9544 \times 2000 \\
 &= 1909 \text{ (approx)}
 \end{aligned}$$



Hence, among 6000 workers, the number of workers getting wages between Rs. 1,100 and Rs. 1,500 is
 $6000 \times 0.3721 = 2232.6 \approx 2233$.

Example 14.36. Assume the mean height of soldiers to be 68.22 inches with a variance of 10.8 inches². How many soldiers in a regiment of 1,000 would you expect to be

(i) over six feet tall, and (ii) below 5.5 feet? Assume heights to be normally distributed.

Solution. Let the variable X denote the height (in inches) of the soldiers. Then we are given :

$$\text{Mean} = \mu = 68.22 \quad \text{and} \quad \text{Variance} = \sigma^2 = 10.8$$

A soldier will be over 6 feet tall if X is greater than 72 (because X is height in inches).

$$\text{When } X = 72, Z = \frac{X - \mu}{\sigma} = \frac{72 - 68.22}{\sqrt{10.8}} = \frac{3.78}{3.286} = 1.15$$

The probability that a soldier is over 6 feet = 72" tall is given by :

$$\begin{aligned} P(X > 72) &= (Z > 1.15) = 0.5 - P(0 \leq Z \leq 1.15) \\ &= 0.5 - 0.3749 = 0.1251 \end{aligned} \quad \text{(From Normal Tables)}$$

Hence, in a regiment of 1,000 soldiers, the number of soldiers over 6 feet tall is :

$$1000 \times 0.1251 = 125.1 \approx 125$$

(ii) The probability that a soldier is below 5.5' = 66" is given by :

$$\begin{aligned} P(X < 66) &= P\left(Z < \frac{66 - 68.22}{\sqrt{10.8}}\right) = P\left(Z < -\frac{2.22}{3.286}\right) \\ &= P(Z < -0.6756) = P(Z > 0.6756) \quad \text{(By symmetry)} \\ &= 0.5 - P(0 < Z < 0.6756) = 0.5 - 0.2501 \quad \text{[From Normal Tables]} \\ &= 0.2499 \text{ (approx.)} \end{aligned}$$

Hence the number of soldiers over 5.5 feet in a regiment of 1,000 soldiers is

$$1000 \times 0.2499 = 249.9 \approx 250.$$

Example 14.37. The average test marks in a particular class is 79. The standard deviation is 5. If the marks are distributed normally, how many students in a class of 200 did not receive marks between 75 and 82? Given :

$$\Pr\{0 \leq Z \leq .7\} = .2580,$$

$$\Pr\{0 \leq Z \leq .8\} = .2881$$

$$\Pr\{0 \leq Z \leq .6\} = .2257,$$

where Z is a standard normal variable.

Solution. If the random variable X denotes the marks obtained by the students in the given test then we are given :

$$X \sim N(\mu, \sigma^2) \quad \text{where} \quad \mu = 79 \quad \text{and} \quad \sigma = 5.$$

The probability that a student gets marks between 75 and 82 is given by :

$$\begin{aligned} P(75 < X < 82) &= P\left(\frac{75 - 79}{5} < Z < \frac{82 - 79}{5}\right) \quad \left(\because Z = \frac{X - \mu}{\sigma} = \frac{X - 79}{5}\right) \\ &= P(-0.8 < Z < 0.6) \\ &= P(-0.8 < Z < 0) + P(0 < Z < 0.6) \\ &= P(0 < Z < 0.8) + P(0 < Z < 0.6) \quad \text{(By symmetry)} \\ &= 0.2881 + 0.2257 = 0.5138 \end{aligned}$$

The probability p that a student does not get marks between 75 and 82 is given by :

$$\begin{aligned} p &= 1 - P(\text{Student gets marks between 75 and 82}) \\ &= 1 - 0.5138 = 0.4862 \end{aligned}$$

Hence, in a class of 200 students, the number of students who did not receive marks between 75 and 82 is given by :

$$200 \times p = 200 \times 0.4862 = 97.24 \approx 97.$$

Example 14.38. The daily wages of 1,000 workmen are normally distributed around a mean of Rs. 70 and with a standard deviation of Rs. 5. Estimate the number of workers whose daily wages will be :

- (i) between Rs. 70 and 72 ; (ii) between Rs. 69 and 72
 (iii) more than Rs. 75 ; (iv) less than Rs. 63 (v) more than Rs. 80.
 (vi) Also estimate the lowest daily wages of the 100 highest paid workers.

[Delhi Univ. B.Com. (Hons.), 1998]

Solution. Let the random variable X denote the daily wages in Rupees. Then X is a normal variable with mean $\mu = 70$ and $\sigma = 5$. The standard normal variable corresponding to X is

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 70}{5}$$

X	63	69	70	72	75	80
$Z = \frac{X - 70}{5}$	$\frac{63 - 70}{5} = -1.4$	-0.2	0	0.4	1	2

... (*)

(i) We want $P(70 < X < 72)$. From Table (*), we get

$$P(70 < X < 72) = P(0 < Z < 0.4) = 0.1554$$

Therefore, the number of workers with daily wages between Rs. 70 and Rs. 72 is :

$$1000 \times 0.1554 = 155.4 \approx 155$$

(ii) $P(69 < X < 72) = P(-0.2 < Z < 0.4)$

[From (*)]

$$= P(-0.2 < Z < 0) + P(0 < Z < 0.4)$$

$$= P(0 < Z < 0.2) + P(0 < Z < 0.4)$$

(By symmetry)

$$= 0.0793 + 0.1554 = 0.2347$$

Hence, the required number of workers is : $1000 \times 0.2347 = 234.7 \approx 235$.

(iii) We want $P(X > 75)$

$$\therefore P(X > 75) = P(Z > 1) \quad [\text{From (*)}]$$

$$= 0.5 - P(0 < Z < 1)$$

[From Fig. 14.14]

$$= 0.5 - 0.3413 = 0.1587$$

Thus, the number of workers with daily wages more than the Rs. 75 is :

$$1000 \times 0.1587 = 158.7 \approx 159$$

(iv) $P(X < 63) = P(Z < -1.4)$ [From (*)]

$$= P(Z > 1.4)$$

[By symmetry, Fig. 14.15]

$$= 0.5 - P(0 < Z < 1.4)$$

$$= 0.5 - 0.4192 = 0.0808.$$

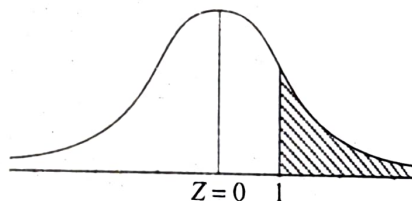


Fig. 14.14

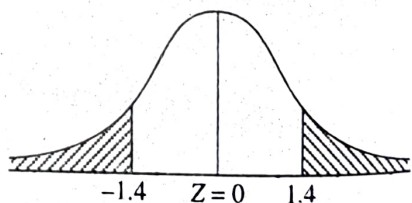


Fig. 14.15

Hence, the number of workers with daily wages less than Rs. 63 is : $1000 \times 0.0808 = 80.8 \approx 81$

(v) $P(X > 80) = P(Z > 2) = 0.5 - P(0 < Z < 2)$

$$= 0.5 - 0.4772 = 0.0228$$

Hence, the number of workers with daily wages over Rs. 80 is : $1000 \times 0.0228 = 22.8 \approx 23$

(vi) Proportion of the 100 highest paid workers is : $\frac{100}{1000} = \frac{1}{10} = 0.10$

We want to determine $X = x_1$, say, such that $P(X > x_1) = 0.10$

$$\text{When } X = x_1, Z = \frac{x_1 - 70}{5} = z_1, (\text{say}).$$

... (**)

Then

$$P(Z > z_1) = 0.10 \Rightarrow P(0 < Z < z_1) = 0.5 - 0.1 = 0.40$$

From the normal probability tables and (**), we get

$$z_1 = \frac{x_1 - 70}{5} = 1.28 \text{ (approx.)} \Rightarrow x_1 = 70 + 5 \times 1.28 = 70 + 6.40 = 76.40$$

Hence, the lowest daily wages of the 100 highest paid workers are Rs. 76.40.

Example 14-39. The income distribution of officers of a certain company was found to follow normal distribution. The average income of an officer was Rs. 15,000.00. The standard deviation of the income of officers was Rs. 5,000.00. If there were 242 officers drawing salary above Rs. 18,500.00, how many officers were there in the company?

[The area under standard normal curve between 0 and 0.7 is 0.2580] [C.A. (Foundation), Nov. 2001]

Solution. Let the r.v. X denote the income (in Rs.) of the officers of the company. Then we are given that :

$$X \sim N(\mu, \sigma^2) \text{ where, } \mu = \text{Rs. } 15,000 \text{ and } \sigma = \text{Rs. } 5,000.$$

Let the number of officers in the company be N . Then, the number of officers in the company drawing salary above Rs. 18,500 is :

$$N \cdot P(X > 18,500) = 242 \text{ (Given).}$$

... (*)

$$\text{When } X = 18,500, Z = \frac{X - \mu}{\sigma} = \frac{18500 - 15000}{5000} = 0.7$$

$$\begin{aligned} \therefore P(X > 18500) &= P(Z > 0.7) \\ &= 0.5 - P(0 \leq Z \leq 0.7) \quad [\text{Fig. 14-16}] \\ &= 0.5 - 0.2580 \\ &= 0.242 \end{aligned}$$

(From Normal Table)

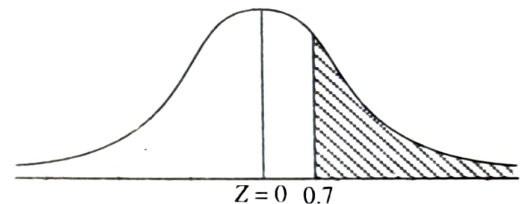


Fig. 14-16

$$\text{Substituting in (*) we get : } N \times 0.242 = 242 \Rightarrow N = \frac{242}{0.242} = 1,000$$

Hence, the number of officers in the company is 1,000.

Example 14-40. Time taken by the crew, of a company, to construct a small bridge is a normal variate

Exponential Distribution

The probability distribution $f(x)$ is given by

$$f(x) = \lambda e^{-\lambda x} \text{ for } x > 0$$

where λ is a parameter

- Mean = $\frac{1}{\lambda} = \mu_1$

- S.D = $\frac{1}{\lambda} = \sigma$

- Variance = $\frac{1}{\lambda^2}$

Q:- The length of telephone conversation is an exponential variate with mean 3 min.

Find the probability that call

(i) ends less than 3 min

(ii) takes between 3 to 5 min.

Sol:- Here Mean = $\frac{1}{\lambda} = 3$

$$\Rightarrow \lambda = \frac{1}{3}$$

$$f(x) = \lambda e^{-\lambda x} = \frac{1}{3} e^{-x/3}$$

$$\begin{aligned} \text{(i)} \quad P(X < 3) &= \int_0^3 f(x) dx = \int_0^3 \frac{1}{3} e^{-x/3} dx \\ &= 1 - \frac{1}{e} \end{aligned}$$

$$\begin{aligned}
 (11) \quad P(3 < X < 5) &= \int_3^5 f(x) dx \\
 &= \int_3^5 \frac{1}{3} e^{-x/3} dx \\
 &= e^{-1} - e^{-5/3}
 \end{aligned}$$

Q:- Find the mean & variance of the exponential distribution
 $f(x) = \frac{1}{b} e^{-(x-a)/b}, x > a$

Sol:- We have $f(x) = \frac{1}{b} e^{-(x-a)/b}, x > a$

Put $x - a = t$
 so that $x > a \Rightarrow t > 0$.

$$f(x) = \frac{1}{b} e^{-t/b}, t > 0.$$

$$\frac{1}{2} =$$

We have $f(x) = \frac{1}{b} e^{-(x-a)/b}, x > a$

$$\begin{aligned}
 \text{Mean} = E(X) &= \int_a^\infty x f(x) dx \\
 &= \int_a^\infty x \frac{1}{b} e^{-(x-a)/b} dx
 \end{aligned}$$

$$= \frac{1}{b} \int_a^{\infty} x e^{-(x-a)/b} dx$$

$$= \frac{1}{b} e^{a/b} \int_a^{\infty} x e^{-x/b} dx$$

$$= \frac{1}{b} e^{a/b} \left[\frac{x e^{-x/b}}{-1/b} - \int \frac{e^{-x/b}}{-1/b} \right]_a^{\infty}$$

$$= \frac{1}{b} e^{a/b} \left[-b x e^{-x/b} + b \int e^{-x/b} dx \right]_a^{\infty}$$

$$= \frac{1}{b} e^{a/b} \left[-x e^{-x/b} + \frac{e^{-x/b}}{-1/b} \right]_a^{\infty}$$

$$= e^{a/b} \left[-x e^{-x/b} - b e^{-x/b} \right]_a^{\infty}$$

$$= e^{a/b} \left[0 - 0 + a e^{-a/b} + b e^{-a/b} \right]$$

$$= e^{a/b} \cdot e^{-a/b} (a+b) = a+b.$$

~~Variance =~~

$$\text{Variance} = E(X^2) - [E(X)]^2$$

$$\therefore E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= \int_a^{\infty} x^2 \frac{1}{b} e^{-(x-a)/b} dx$$

$$= \frac{1}{b} e^{a/b} \int_a^{\infty} x^2 e^{-x/b} dx$$

$$\begin{aligned}
 &= \frac{1}{b} e^{a/b} \left[\frac{x^2 e^{-x/b}}{-1/b} - \int \frac{2x e^{-x/b}}{-1/b} dx \right]_a^\infty \\
 &= \frac{1}{b} e^{a/b} \cdot b \left[-x^2 e^{-x/b} + 2b \int x e^{-x/b} dx \right]_a^\infty \\
 &= e^{a/b} \left[-x^2 e^{-x/b} + \frac{2bx e^{-x/b}}{-1/b} - 2b \int \frac{e^{-x/b}}{-1/b} dx \right]_a^\infty
 \end{aligned}$$

$$\begin{aligned}
 &= e^{a/b} \left[-x^2 e^{-x/b} - 2b^2 x e^{-x/b} + 2b^2 \frac{e^{-x/b}}{-1/b} \right]_a^\infty \\
 &= e^{a/b} \left[-x^2 e^{-x/b} - 2b^2 x e^{-x/b} - 2b^3 e^{-x/b} \right]_a^\infty \\
 &= e^{a/b} \left[0 - 0 - 0 + a^2 e^{-a/b} + 2ab^2 e^{-a/b} + 2b^3 e^{-a/b} \right]
 \end{aligned}$$

$$\begin{aligned}
 &= e^{a/b} \cdot e^{-a/b} \left[a^2 + 2ab^2 + 2b^3 \right] \\
 &= a^2 + 2ab^2 + 2b^3
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Variance} &= E(X^2) - [E(X)]^2 \\
 &= a^2 + 2ab + 2b^3 - (a+b)^2 \\
 &= \cancel{a^2} + \cancel{2ab} + 2b^3 - \cancel{a^2} - \cancel{2ab} - b^2 \\
 &= 2b^3 - b^2
 \end{aligned}$$